Design high availability architecture

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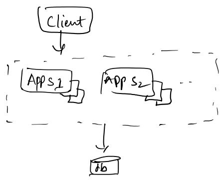
Design High Availability Architecture

This can be asked in many forms like
- Design Data Resiliency Architecture [Resiliency means something which has the cability to comes out of failure]
- Design Architecture to achieve 99.999% Availability [Also called 5 9's = 99.999]
- Design to avoid Single point of failure
- Active/Passive v/s Active/Active Architecture

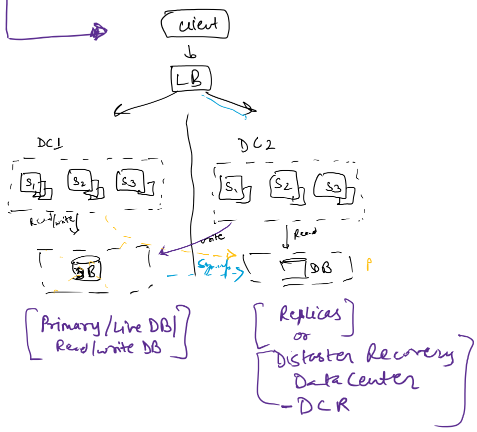
Single Node



if DB goes down, everything collapse

MultiNode

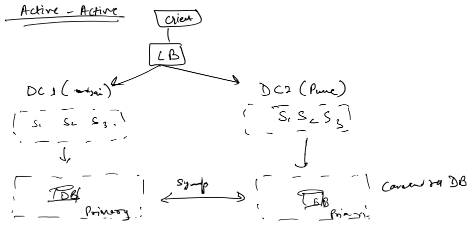
- Active Passive Architecture
- Active Active Architecture



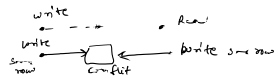
- Req Comes in server fails then DCR uses Primary DB we can not use Replica DB because Oracle/Mysql Postgres are not multi master. means they can only write in only one primary DB
- write (Post) comes than it trasfer to Live DB
- Read (Get) comes then it can be read from Replica DB
- when Primary DB fails than all Req goes to Replica DB and Replica DB becomes primary DB. and when primary DB up than it becomes Replica

- Diadvantage :
  - Latency comes when Req comes to Replica
  - When primary is switching than there is gap of time untill connection is made till that time write operation will fail.
  - Too much write operations

Active - Active



- Both DB will be utlized
- Syncup is complex because



- Good thing is we are utlizing Resources in our data centers so more traffic can be hadled